

Business Proposal Letter

January 7, 2026

TO: THE SCIENCE FACULTY

Southcom National High School (SNHS)

SUBJECT: Proposal for Digital Laboratory Transformation & Pilot Testing Invitation

Dear Fellow Science Educators,

As an alumnus of **Southcom National High School**, I am well aware of the daily challenges we face in our laboratories: the missing beakers that never get replaced, the struggle of identifying "freeloaders" in group work, and the hours wasted on manual logbooks.

I am proposing a partnership to implement a **Hybrid Laboratory Inventory & Collaborative LMS**, a system I have developed specifically to solve these manual lab headaches through automation, real-time tracking, and data-driven accountability.

1. The Problem-Solution Statement

The current reliance on manual processes in the laboratory setting leads to significant financial loss and administrative burden. The proposed system directly addresses these issues through digital innovation.

The Manual Struggle	The Digital Solution
Asset Leakage: ₱30k–₱50k of equipment is lost or broken annually due to paper logbooks that fail to enforce individual accountability.	Distributed Liability: QR-based borrowing that links every item to the specific student holding it, not just the group leader.
Grading Guesswork: Teachers struggle to prove who actually worked in a group and who simply "signed the paper."	Forensic Heatmaps: A dashboard showing the exact percentage of effort, word count, and time contributed by each student.

The Manual Struggle	The Digital Solution
Administrative Burnout: Teachers spend 8–17 hours/week on non-teaching tasks (inventory updates, paper reports).	Process Automation: Instant stock updates and digital submission loops that reduce administrative work by an estimated 60%.

2. System Workflow: The Smart-Lab Engine

The system is designed as a complete **Laboratory Ecosystem** that handles both the physical logistics of equipment and the academic output of students.

A. Logistics and Inventory Engine

- **The Borrowing Wizard:** Students select apparatus via a "Shopping Cart" interface. The system performs an instant inventory check against `Available_Qty` and generates a digital Borrowing Slip with a QR Code.
- **Approval Workflow:** Teachers act as gatekeepers, reviewing student requests via a digital dashboard to approve or deny them before items are released.
- **Liability Management:** Students upload visual proof (photos) of returned or broken items. Liabilities are tied to individual `UserIDs` within a `SessionID`, ensuring the correct student is held accountable for school assets.

B. Collaborative Workspace and Forensic Engine

- **Section-Based Editing:** Reports are broken into modular cards (Introduction, Methodology, Results, etc.).
- **Concurrency Locking:** I have implemented a custom **Lock Manager**. When one student opens a section, it locks for all others in real-time to prevent overwriting conflicts.
- **The Forensic Engine:** The system calculates "Net Contribution" (characters added vs. deleted). It includes **Paste Detection** to flag text injections (e.g., >500 characters in one second), alerting teachers to potential AI or Wikipedia copy-pasting.
- **Revision Workflow:** Teachers can "Return" work for revision. This triggers a state change that unlocks the workspace and displays a "Revision Needed" banner with instructor feedback.

3. Flexible Deployment Options

The system offers two deployment options based on the department's needs and resource availability:

Option	How it Works	Requirements/Notes
Option 1: Local-Host Setup (No Internet Required)	A faculty member's personal laptop can be used as the central host. Students connect to the laptop's signal via Wi-Fi.	Works offline , ensuring 100% uptime regardless of internet stability.
Option 2: Online Cloud Hosting	Hosted on a high-speed Singapore-based server for access anywhere (school or home).	Requires an annual hosting/domain fee (approx. ₱7,500) to be covered by the department.

4. Pricing & Professional Commitment

I am currently a **Second-Year student graduating from the Associate in Computer Technology (ACT) program, majoring in Application Development**. Because this is my **first-ever professional business deal**, I am offering my services at a rate significantly below standard market value to build my portfolio while giving back to my Alma Mater.

The investment required is detailed in the table below:

Service Component	Corporate Agency Rate	SNHS Alumnus Rate
Custom PHP/MySQL Engine	₱150,000+	₱15,000
QR & Lock-Manager API	₱40,000+	INCLUDED
Data Digitization (Labor)	₱15,000	INCLUDED
TOTAL INVESTMENT	₱205,000+	₱15,000

5. CALL TO ACTION: THE 1-DAY BETA TEST

I am requesting a Safety Check to prove the system's value in a live environment before any commitment is made. This session will serve as a full tutorial and guided walkthrough of the platform's capabilities.

Pilot Testing Details:

Detail	Information
Date:	January 12, 2026 (Monday)
Target Class:	11 STEM
Activity:	Full System Tutorial and Guided Walkthrough

I will be on-site to handle the local server setup using a personal laptop to host the session. During this walkthrough, I will guide both faculty and students through the end-to-end process, from generating QR slips to analyzing forensic contribution heatmaps. Your only role is to observe how the system manages your students and provides you with objective grading data in real-time.

Are you ready to witness the transformation of the SNHS laboratory this coming Monday?

Sincerely,

mark ando

ACT - Application Development | SNHS Alumnus